Helping Hand Release Candidate

M5 Milestone Work Outline

Course: CEN 4908C

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Work Completed

- **Nine-axis IMU design and firmware:** We changed our planned IMU from the current six-axis unit to the Adafruit ICM-20948 nine-axis breakout, which adds a three-axis magnetometer. We updated the firmware to detect and initialize the new IMU, read its accelerometer and gyroscope, and read the AK09916 magnetometer. The older MPU-style sensor is still supported. Magnetic readings are available through serial diagnostics; the current BLE motion fields and word model still use accelerometer/gyroscope data. Installing the new board, calibration, and orientation correction are still pending.
- **Piezo control code:** We added pin setup and timed PWM control for all ten piezo buzzers: one on each finger and five in an X on the back of the hand. All outputs start off, and the code can play one buzzer at a time without blocking the sensor/BLE loop. The firmware also checks for repeated pins or conflicts with the flex sensors and I2C connections. Connecting the buzzers to automatic finger corrections and testing the driver circuit are still pending.
- **PCB schematic and layout planning:** We continued the KiCad schematic for the ESP32-S3 and 15 resistors used in the five flex-sensor and ten piezo circuits. We have also been planning the board layout. Placement, routing, fabrication, and testing of the finished PCB are not complete.
- **Glove layout and sewing preparation:** We planned the placement of one IMU, five flex sensors, and ten piezos, and practiced making small circuits with conductive thread. The proposed layout puts a flex sensor and piezo at each finger, the IMU on a secured mount on the back of the hand, and five piezos in the dorsal X pattern. We plan to secure components with nonconductive stitching and use fabric channels so the flex strips can bend without pulling on their terminals. Sewing the full assembly is still pending.
- **Fabric insulation plan:** We proposed five fabric layers total, including the glove liner: inner liner, first routing fabric, insulating separator, second routing fabric, and outer cover. The two routing levels let us separate connections where they cross. Each routing fabric will be sewn separately so conductive stitches do not pass through the separator into the other level. Terminals, knots, and extra crossings still need local insulation. Five layers is a starting design, not a tested minimum or a guarantee against shorts; the final wiring layout may need adjustments. We plan to use insulated flexible wire for IMU I2C and power connections and check continuity and shorts with the glove flat and bent before powering it.
- **Flex-sensor replacement planning:** About two flex sensors were reading bent when straight. We ordered five Adafruit Short Flex Sensor #1070 replacements. Repair and recalibration are needed before collecting reliable new recordings; installation has not been confirmed.

- **TCN improvement review:** We reviewed ways to improve the existing word model, including collecting more repetitions across wearers and wrist angles, retaining more of each complete gesture, and rejecting idle or unrelated movement. These are proposed changes, not a newly trained model or a measured accuracy improvement. Hello, Please, and Yes recognition is carried forward from M4.
- **Firmware checks:** Host-side simulated-I2C checks passed for the new IMU's register setup, signed sensor values and units, magnetometer readiness and overflow handling, reconnect behavior, the older sensor path, and address 0x69. These are software checks. The full ESP32 build attempt timed out while PlatformIO tried to install a missing tool; board compilation and physical testing of the new IMU are not confirmed.
- **Release documentation:** We updated the release-candidate README with the M5 rubric table, current hardware plans, piezo pin assignments, and the work still needing testing. The existing account flow, saved progress, dedicated practice pages, and three-word TCN are retained from the beta build rather than counted as new M5 features. New-board bring-up, buzzer output, sewn-glove checks, and current glove-to-phone testing are **Not run**.